

An Observability Requirements Framework for Generative AI Systems in the Public Sector

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Abstract

Generative artificial intelligence is moving from experimental pilots to production-facing services across the Singapore public sector, supported by a sustained whole-of-government effort that includes the National AI Strategy 2.0, the Smart Nation 2.0 vision, and central platforms developed by the Government Technology Agency. As deployments scale, agencies face a systemic observability gap: telemetry practices inherited from earlier machine-learning operations do not capture the artefacts that matter for generative AI oversight, including reasoning traces, tool invocations, retrieval provenance, refusal events, and groundedness signals. This paper introduces the Observability Requirements Framework, a structured, risk-tiered specification that organises telemetry into six operational categories spanning agent-level observability, agent performance, user interaction, system health, safety and security compliance, and product lifecycle governance. The framework pairs the requirements table with mathematical metric definitions to enable consistent implementation across government agencies with differing technical maturity. The framework is designed to align with the principles articulated in Singapore’s Model AI Governance Framework for Generative AI, the Model AI Governance Framework for Agentic AI, and the Agentic Risk and Capability framework released by the Government Technology Agency (GovTech). We position the framework as the evidentiary substrate that connects high-level governance principles to engineering practice. A companion manuscript will demonstrate the framework on two worked examples and outline its integration into central platform products and policy implications within GovTech.

Keywords: LLMOps, AgentOps, observability, public sector AI, Singapore, AI governance, agentic systems, responsible AI.

1 Introduction

Generative artificial intelligence has matured from research demonstrations into production infrastructure, and the public sector is no longer a bystander to that transition. Singapore has been particularly explicit about this shift. The National AI Strategy 2.0, launched in December 2023, repositioned artificial intelligence as a national necessity rather than an opportunity ([Smart Nation and Digital Government Office, 2023](#)), and the Smart Nation 2.0 vision unveiled in October 2024 framed trust, growth, and community as the goals that should guide the country’s digital transformation ([Ministry of Digital Development and Information, 2024](#)). In parallel, the Government Technology Agency of Singapore (GovTech), the technology arm of the Singapore government, has rolled out a suite of central platform products that put generative AI directly into the hands of public officers, including the Pair productivity suite ([Government Technology Agency of Singapore, 2024b](#); [United Nations Development Programme and Open Government Products, 2024](#)), the AIBots platform for retrieval-augmented chatbots ([Government Technology Agency of Singapore, 2025c](#)), and the LaunchPad innovation environment ([Government Technology Agency of Singapore, 2025e](#); [Microsoft Customer Stories, 2024](#)).

This rapid uptake creates an operational problem that has not yet been adequately addressed. Public sector deployments of generative AI must answer to citizens, regulators, parliamentary oversight, and internal audit functions, all of which expect detailed evidence about how a system has behaved over time. Agencies are accountable not only for what their AI systems do but also for what those systems chose not to do, what alternatives were considered, what data was retrieved, and which guardrails were activated.

The artefacts that matter for this kind of oversight, namely reasoning traces, tool invocations, retrieval provenance, refusal events, and groundedness signals, are precisely the artefacts that pre-generative monitoring practices were not designed to capture. Without a structured specification, observability tends to be ad hoc, inconsistent across agencies, and difficult to align with governance frameworks.

The need for such a specification has become more acute as agentic AI moves into production. Agentic systems plan across multiple steps, invoke tools, and act on the world, and the consequences of their actions can be irreversible (Khoo et al., 2025; Dong et al., 2024). The Infocomm Media Development Authority (IMDA) recognised this trajectory by releasing the Model AI Governance Framework for Agentic AI in January 2026, which became the world’s first comprehensive governance framework targeted specifically at agentic deployments (Infocomm Media Development Authority, 2026; MDDI press release, 2026). At the technical layer, the GovTech complemented this policy direction with the Agentic Risk and Capability (ARC) framework, an open-source technical governance framework that maps agentic capabilities to risks and recommended controls (Khoo et al., 2025; Government Technology Agency of Singapore, 2025d). Both frameworks call for monitoring, traceability, and tamper-evident audit trails, but they stop short of specifying what telemetry agencies should collect, how to prioritise it under resource constraints, and how to define the metrics that flow from it.

This paper closes that gap by introducing the Observability Requirements Framework for generative AI systems in the public sector. The framework is risk-tiered, implementation-agnostic, and designed to be adopted by government agencies operating at very different levels of technical maturity. It organises telemetry into six operational categories that jointly cover the artefacts necessary to reconstruct, evaluate, and govern an agentic system, and it pairs each category with a table of metrics, logs, and traces, mathematical definitions of each metric, and a four-tier requirements matrix that maps priority to system risk level. The framework is intended to align with existing governance principles, in particular those articulated in the Model AI Governance Framework for Generative AI (Infocomm Media Development Authority and AI Verify Foundation, 2024), the Model AI Governance Framework for Agentic AI (Infocomm Media Development Authority, 2026), and the ARC framework (Khoo et al., 2025; Government Technology Agency of Singapore, 2025d).

The contribution of this paper is threefold. **For the research community**, the framework provides the first publicly-released, risk-tiered telemetry specification for generative and agentic AI systems in a national government, and is intended as a reusable artefact that researchers can extend, critique, and benchmark against deployments in other jurisdictions. By specifying telemetry in a stack-aware way that cuts across LLMOps, RAGOps, and AgentOps, the framework creates the data substrate on which observability into real-world generative AI behaviour (e.g., hallucination rates in production, handover dynamics between human and agent, fairness across user segments, drift over model upgrades) can be conducted with comparable measurements across agencies and over time. **For the practitioner community**, the framework gives engineering and platform teams a concrete checklist of what to log, what to measure, and how to prioritise, that is implementable on open-source stacks and aligns with the governance frameworks under which they operate. **For the general public and civil society**, the framework makes explicit what evidence a public-sector AI system should be able to produce about its own behaviour. By specifying telemetry as a requirement rather than a feature, the framework supports public scrutiny, informed parliamentary oversight, and freedom-of-information style queries on the operation of consequential AI services, and it gives citizens a vocabulary for asking how a system behaved, what it considered, and why.

The remainder of this paper is structured as follows. Section 2 introduces the operational disciplines of LLMOps, RAGOps, and AgentOps, explains why generative AI operations are qualitatively harder than classical software operations, and positions the framework within the emerging full-stack view of AI observability. Section 3 situates the work in Singapore’s national push on generative AI and describes GovTech’s role as the central technology agency. Section 4 surveys publicly disclosed Singapore government generative AI deployments. Section 5 examines the considerations that distinguish public sector deployment from commercial practice. Section 6 argues that observability is the substrate that connects governance principles to engineering practice. Section 7 presents the Observability Requirements

Framework in full, with its goals, adoption flow, components, and the supporting tables. Section 8 closes the present manuscript and previews a companion paper that will apply the framework to two worked examples and discuss its integration into central platform products and its policy implications within GovTech.

2 Background: LLMOps, RAGOps, and AgentOps

2.1 From MLOps to Generative AI Operations

Machine Learning Operations (MLOps) emerged from the recognition that putting machine learning models into production requires the same discipline that DevOps brought to software, with additional concerns around data versioning, model drift, and reproducibility. The discipline grew up around discriminative models with structured inputs, deterministic evaluation regimes, and well-understood failure modes, and its tooling reflects that history. Generative AI introduces a different set of problems: open-ended outputs that resist reference-based evaluation, prompts that function as a primary iteration surface, and emergent behaviour that depends on interactions between models, retrieval pipelines, and tool calls (Pahune and Akhtar, 2025; Diaz-De-Arcaya et al., 2024). These differences are not incremental, and they have given rise to a hierarchy of derivative disciplines, often described as MLOps subsuming GenAIOps, which in turn subsumes LLMOps, with RAGOps as a specialised subset and AgentOps as a sibling discipline focused on autonomous multi-step execution.

2.2 LLMOps: Large Language Model Operations

Large Language Model Operations (LLMOps) covers the practices, tools, and processes for managing LLM-based applications across their full operational life, from initial deployment through continual adaptation and eventual deprecation (Pahune and Akhtar, 2025; Diaz-De-Arcaya et al., 2024). Two characteristics distinguish LLMOps from earlier MLOps practice. First, prompt versioning replaces feature engineering as a primary iteration surface, which means that practitioners need infrastructure to version, test, and roll back prompts in the same way they version code (Langfuse, 2024). Second, monitoring must accommodate non-deterministic outputs and unstructured text, which makes traditional reference-based metrics inadequate; teams instead rely on a combination of structured logging, sampling-based human review, and automated evaluation methods such as LLM-as-a-judge (Huang et al., 2025). We treat these concerns under two complementary headings: the LLM lifecycle and LLM maintenance.

2.2.1 The LLM Lifecycle

The LLM lifecycle covers the discrete phases through which an LLM-based application moves between conception and retirement. It begins with model selection and prompt design, proceeds through evaluation against representative inputs, fine-tuning where appropriate, safety and red-team review, and pre-deployment sign-off. After deployment, the lifecycle continues through inference serving with capacity management, safety guardrails layered in front of and behind the model, and online evaluation that samples production traffic for quality and safety assessment. The lifecycle ends with controlled deprecation, including rollback paths and a planned migration when an underlying model version is retired by its provider. This lifecycle inherits the discipline of MLOps but adds artefacts that have no clean MLOps analogue, including the prompt itself, the system message, the choice of decoding parameters, and the LLM-as-judge evaluators used to score open-ended output.

2.2.2 LLM Maintenance and Continual Adaptation

Maintenance is the part of LLMOps that runs continuously after deployment and is, in practice, where the largest share of operational effort accumulates. It comprises four recurring activities. **Model refresh and migration** addresses the fact that closed-weight LLMs are routinely updated, deprecated, or replaced by their providers, often with limited notice; production prompts that worked reliably against one model

version can regress against the next, and maintaining a deployment therefore means continuously re-validating prompts against new model versions and managing the migration when a model is sunset (Pahune and Akhtar, 2025). **Prompt and policy maintenance** covers the routine revision of system prompts, retrieval rubrics, refusal policies, and tool-use instructions as the surrounding policy environment changes, as new failure modes are discovered, and as user behaviour evolves; each revision needs to be versioned, evaluated against a regression suite, and rolled out with the same care as a code change. **Drift remediation** covers the detection and response to changes in the distribution of inputs (new user intents, novel topics, adversarial probing), changes in the distribution of outputs (hallucination rate, refusal rate, sentiment shift), and changes in the upstream retrieval corpus or embedding model that perturb the grounded behaviour of the system. **Continual evaluation** covers the standing infrastructure that runs offline regression batteries, samples online traffic, computes safety and quality metrics on a recurring schedule, and triggers human-in-the-loop review when thresholds are breached. The boundary between maintenance and incident response is therefore deliberately porous: maintenance is what catches the drift before it becomes an incident, and the observability substrate this paper specifies is what makes that detection possible.

2.3 RAGOps: Retrieval-Augmented Generation Operations

Retrieval-Augmented Generation (RAG) extends an LLM with the ability to retrieve external documents at inference time, grounding generation in evidence drawn from a knowledge base (Lewis et al., 2020). RAG has become the dominant pattern for enterprise LLM deployments, with industry surveys reporting that approximately 60% of enterprise LLM systems now incorporate retrieval (Xu et al., 2025). RAGOps is the operational discipline that governs the additional surfaces that retrieval introduces: ingestion pipelines, chunking strategies, embedding model selection, vector index maintenance, retrieval evaluation, and content provenance (Xu et al., 2025). For the public sector, RAGOps is particularly consequential because government knowledge bases evolve continuously as policies, regulations, and service descriptions change. The freshness of the retrieval corpus is therefore a first-order operational concern, and the boundary between a stale knowledge base and an inaccurate response is direct and measurable. The same lifecycle-versus-maintenance distinction applies to RAGOps as to LLMOps, with the maintenance side dominated by index rebuilds, embedding-model migrations, and continuous evaluation of retrieval quality.

2.4 AgentOps: Agent Operations

Agent Operations (AgentOps) is the most recent and arguably the most demanding of the three disciplines. It addresses the operational concerns of autonomous and semi-autonomous agent systems that plan, decompose, and execute tasks using tools and memory (Dong et al., 2024). Where LLMOps focuses on the model and RAGOps focuses on the data, AgentOps focuses on the trajectory of execution: planning traces, tool invocations, memory reads and writes, subtask decompositions, and retry lineages. Capturing these artefacts end-to-end is necessary because agent failures are often not visible at the level of any single LLM call. A miscoordinated multi-agent handoff, a runaway tool invocation loop, or a silently-corrupted memory write can produce a defective outcome even when each individual component appears to behave correctly (Dong et al., 2024; Lee and Tiwari, 2024).

The emerging taxonomy of AgentOps tools and artefacts (Dong et al., 2024) and parallel work on Agentic RAG architectures (Singh et al., 2025) confirm that observability is not a peripheral concern for agentic systems but a structural requirement. The OpenTelemetry community has responded by working on dedicated semantic conventions for generative AI agents, including attributes for tracing tasks, actions, agent teams, artifacts, and memory (OpenTelemetry Project, 2025a,b). These conventions are still maturing, but they signal an industry-wide convergence on the conclusion that agentic observability requires its own vocabulary. As with LLMOps, the operational discipline divides naturally into a lifecycle component (agent design, evaluation, deployment, deprecation) and a maintenance component (tool catalogue updates, prompt and policy revision, continuous trajectory evaluation, and adaptation to changes

in the underlying LLMs and retrieval layers).

2.5 Why AI Operations Are Harder than Software Operations

LLMOps, RAGOps, and AgentOps are sometimes treated as DevOps with a model in the loop. That framing understates the difficulty. Five properties of LLM-based systems make their operational discipline qualitatively different from, and in important respects more demanding than, classical software engineering and DevOps practice.

Non-determinism is structural, not a bug. Classical software returns the same output for the same input and state, which is the property on which unit testing, regression testing, idempotency analysis, and crash-equivalence checking all rest. On the other hand, LLMs return different outputs for the same input across different decoding draws, and even temperature-zero decoding can drift when the underlying model version is updated. This pulls the rug out from under reference-based evaluation and forces the operational discipline to reason about output distributions rather than individual outputs, with statistical methods (sampling-based human review, LLM-as-a-judge, drift detection) replacing exact equality (Huang et al., 2025; Comet, 2026).

State is distributed across artefact classes that DevOps does not recognise. A traditional service's state lives in code, configuration, and database rows. An LLM-based service's state additionally lives in the model version, prompt, system message, retrieval index, embedding model that built the index, tool definitions, the agent's working memory, and conversation history. Each of these is a versioned artefact that can change independently and produce silent behavioural shifts. Operations have to extend their notion of "what is deployed" across all of these surfaces.

Failure modes are emergent and often silent. A latent prompt regression on a model upgrade, an embedding drift after an index rebuild, an indirect prompt injection through retrieved content, or a multi-step tool-use loop that quietly burns budget without raising an exception. None of these is detected by green health checks and HTTP 200 status codes (Comet, 2026; Dong et al., 2024). The signals that matter are semantic (was the answer faithful to the retrieved evidence? did the agent escalate at the right point?) and behavioural (did refusal rate spike for a particular user segment? did the rate of human override jump after the last prompt change?), and these signals only become visible if the observability stack is designed to capture them.

The adversarial surface has no classical analogue. Prompt injection, jailbreaks, indirect injection via retrieved content, and tool-use exploitation are all unique to LLM-based systems. The threat model is not a buffer overflow or an SQL injection but an open-ended natural-language attack against a probabilistic decision-maker, and the defences are partial. This means the security telemetry needed in production goes well beyond authentication logs and WAF events; it has to capture prompt content, retrieved context, refusal decisions, and the trajectory by which the system reached a contested action (Lee and Tiwari, 2024; Dong et al., 2024).

Accountability obligations exceed those of software. Public-sector users of generative AI are answerable to citizens, regulators, parliamentary oversight, and internal audit for reasoning processes, not only for outcomes. A correct answer reached through a faulty retrieval and a hallucinated intermediate step may still be a governance failure. Software DevOps does not normally need to defend the internal trajectory of a request; AI operations routinely do.

Together these properties mean that LLM-based systems require an operational discipline whose unit of observation is the trace of a reasoning process, not the response of a service. Treating LLMOps as DevOps with a model attached produces observability stacks that are blind to the failure modes that matter.

2.6 AI Observability and the Full-Stack View

The industry response to these difficulties has converged on the concept of *AI observability*: the practice of collecting, analysing, and correlating telemetry across the full AI application stack so that the semantic, behavioural, and agentic properties of the system are visible alongside its computational properties

(Comet, 2026; Dynatrace, 2026; MLflow, 2026). AI observability is best understood not as a new discipline in opposition to LLMOps, RAGOps, or AgentOps, but as the integrative concern that cuts horizontally across them. Where LLMOps governs the model layer, RAGOps governs the data and retrieval layer, and AgentOps governs the execution-trajectory layer, AI observability is the substrate that connects all three by extending the classical three pillars of observability — metrics, logs, and traces — to the artefacts that LLM-based systems actually produce.

In practice, that extension has three dimensions. **Computational observability**, inherited from DevOps and SRE, tracks the unit economics and infrastructure health of the system: latency, throughput, error rates, token usage, cost per session (Comet, 2026). **Semantic observability** tracks the meaning of what the system produces: prompt and response quality, retrieval relevance, groundedness of generated content, refusal events, hallucination rate, and safety-policy violations. **Agentic observability** tracks the trajectory by which the system reached its output: planning steps, tool invocations, memory operations, subtask decompositions, retry lineage, and end-to-end traces across multi-agent handoffs (Dong et al., 2024; OpenTelemetry Project, 2025a).

The industry has begun to standardise on this full-stack view. The OpenTelemetry community has been developing semantic conventions for generative AI agents that extend the standard span and event vocabulary with attributes for prompts, completions, tool calls, agent invocations, and memory operations (OpenTelemetry Project, 2025a,b). Open-source platforms including Langfuse and Arize Phoenix have built production-grade tracing, prompt management, and evaluation stacks that organise around this extended trace model (Langfuse, 2024; Arize AI, 2024), and a growing ecosystem of enterprise vendors offers managed AI observability with comparable surface area (Dynatrace, 2026; MLflow, 2026). The Observability Requirements Framework introduced in this paper sits within this emerging consensus. It does not invent a new observability stack; it specifies what telemetry needs to be captured by whatever stack an agency adopts, organised around the operational concerns that matter for public-sector deployment.

2.7 Positioning the Observability Requirements Framework

Within this landscape, the Observability Requirements Framework is positioned not as a competitor to LLMOps, RAGOps, or AgentOps, but as a concrete specification of the AI observability concerns common to all three when they are exercised together. Most public sector generative AI deployments combine an LLM core, a retrieval layer over a government knowledge base, and at least some degree of agentic execution; the framework therefore specifies telemetry across all three layers in a single artefact, rather than asking practitioners to assemble three separate specifications. The next four sections situate this artefact in the Singapore policy context, examine public-sector deployment specifics, and present the framework in full.

3 Singapore’s Push Towards Generative AI Adoption in Government

3.1 National Strategy and Generative AI as a Public-Sector Priority

Singapore was among the first countries to articulate a national AI strategy, publishing its first edition in 2019 and a refreshed second edition, the National AI Strategy 2.0, in December 2023 (Smart Nation and Digital Government Office, 2023; Centre for Technology, Robotics, Artificial Intelligence & the Law (TRAIL), 2024). The strategy is organised around the slogan “AI for the Public Good, for Singapore and the World” and frames artificial intelligence as a transformative capability that the country must master rather than merely adopt. Among its fifteen actions, the strategy explicitly calls for accelerating public sector adoption of AI (Smart Nation and Digital Government Office, 2023), an objective that has been further reinforced in the Smart Nation 2.0 vision launched by Prime Minister Lawrence Wong in October 2024 (Ministry of Digital Development and Information, 2024). The Smart Nation 2.0 vision identifies trust, growth, and community as the three pillars of Singapore’s continuing digital transformation, and

includes a S\$120 million fund for AI for Science alongside broader initiatives in digital infrastructure and online safety.

Singapore has paired this developmental orientation with an unusually proactive governance posture. The Personal Data Protection Commission (PDPC) and the Infocomm Media Development Authority (IMDA) released the second edition of the Model AI Governance Framework in 2020 (Personal Data Protection Commission, 2023), the Model AI Governance Framework for Generative AI in May 2024 (Infocomm Media Development Authority and AI Verify Foundation, 2024), and the Model AI Governance Framework for Agentic AI in January 2026 (Infocomm Media Development Authority, 2026; MDDI press release, 2026). The Generative AI framework articulates nine dimensions, ranging from accountability and explainability to content provenance and AI for public good, and the Agentic AI framework adds four further dimensions: assessing and bounding risks upfront, making humans meaningfully accountable, implementing technical controls and processes, and enabling end-user responsibility (Infocomm Media Development Authority, 2026). AI Verify, the world's first AI governance testing framework and toolkit, was launched as a minimum viable product in May 2022 and is now stewarded by the AI Verify Foundation (Personal Data Protection Commission and Infocomm Media Development Authority, 2022; Infocomm Media Development Authority, 2023). The Personal Data Protection Act 2012 provides the legal substrate for personal data handling (Government of Singapore, 2012), and was supplemented in March 2024 by Advisory Guidelines on the Use of Personal Data in AI Recommendation and Decision Systems (Personal Data Protection Commission, 2024). Together, these instruments have placed Singapore among the small number of jurisdictions to articulate structured, technically detailed governance guidance for advanced AI systems (Allen et al., 2025).

3.2 GovTech as the Central Technology Agency

GovTech was established in 2016 to spearhead the development and implementation of digital solutions for government, citizens, and businesses (Government Technology Agency of Singapore, 2024a). As the central technology agency, GovTech occupies a position that is closer to a platform operator than a traditional ministry. It builds shared digital infrastructure, including the Singpass digital identity (Perdana and Mokhtar, 2025), runs cross-agency innovation programmes, and provides technical leadership across the public service. In the generative AI era, this central role has become particularly visible. GovTech and the Open Government Products team have rolled out a portfolio of central AI platforms intended to make generative AI safely and uniformly available across the whole-of-government (Leong, 2024; Civil Service College Singapore, 2024).

GovTech's role goes beyond product development, and has also taken the lead in articulating how agentic AI should be governed at a technical level. The Agentic Risk and Capability (ARC) framework, released by GovTech's Responsible AI team in 2025 (Khoo et al., 2025; Government Technology Agency of Singapore, 2025d), provides a hierarchical capability taxonomy for agentic systems, distinguishes component, design, and capability-specific risks, and offers technical control specifications with explicit risk-to-control mappings. ARC is one of the most concrete technical governance frameworks publicly available for agentic AI, and it is positioned as a complement rather than a substitute for the policy-level guidance issued by IMDA and PDPC (Government Technology Agency of Singapore, 2025b,a). The Observability Requirements Framework introduced in this paper is intended to complement ARC at the telemetry layer: where ARC tells agencies which controls to put in place, the Observability Requirements Framework tells them what to log, what to measure, and how to report.

GovTech also operates the central observability platform on which agencies build their production monitoring practice. StackOps is the whole-of-government managed monitoring service offered by GovTech as a SaaS platform on Government on Commercial Cloud, approved for workloads up to the Confidential Cloud-Eligible classification and aligned with IM8 standards (Government Technology Agency of Singapore, 2026c,b). The platform collects logs, traces, and metrics from applications, microservices, servers, and databases, supports over two hundred pre-built integrations spanning containers, Kubernetes pods, and serverless components, and ships with Application Performance Monitoring, Synthetic Monitoring for end-user experience, and a Security Information and Event Management capability

(Government Technology Agency of Singapore, 2026a). StackOps also implements a SIEM-rule in-sourcing model and an extended data schema for sharing monitoring data across agencies, which positions it as the operational substrate on which whole-of-government observability standards can be implemented in practice (Government Technology Agency of Singapore, 2026c).

3.3 From Pilots to Production

The maturation arc within Singapore's public sector tracks a familiar pattern. Early generative AI deployments start off as internal productivity pilots, including the first iteration of Pair as an internal ChatGPT alternative for public officers (United Nations Development Programme and Open Government Products, 2024). As experience accumulated, agencies moved to higher-stakes deployments, including public-facing chatbots, retrieval-augmented domain assistants built on AIBots (Government Technology Agency of Singapore, 2025c), and most recently agentic systems running on air-gapped Google Distributed Cloud infrastructure (GovInsider, 2025). The number and diversity of deployments now in production means that the cost of an inconsistent or under-specified observability practice is substantial. A common, risk-tiered framework that scales across this spectrum is therefore not a future requirement but a present one.

4 Public Sector Use Cases of Generative AI

The Singapore public sector has accumulated a portfolio of publicly disclosed generative AI deployments that span internal productivity, citizen-facing services, and sector-specific applications. This section summarises the most prominent of these use cases.

4.1 Internal Productivity and Knowledge Tools

The Pair suite is the most widely deployed internal productivity tool in the Singapore public sector. Pair Chat is a secure, government-tailored alternative to ChatGPT, available to all public officers on government-issued devices (Government Technology Agency of Singapore, 2024b). Within the first two months of its trial launch in mid-2023, Pair attracted more than eleven thousand users across more than one hundred agencies (Government Technology Agency of Singapore, 2024b), and by mid-2024 the platform had over sixty thousand registered users and more than ten million queries answered (United Nations Development Programme and Open Government Products, 2024). By 2025, approximately fifty thousand public officers were reported to be using Pair to build customised assistants and enhance productivity (GovInsider, 2025). Pair Noms, an AI-assisted meeting-minutes tool, and Pair Search, a policy-relevant search tool over Hansard parliamentary reports, Supreme Court judgments, and legislation, extend the suite to specific public-service workflows (Government Technology Agency of Singapore, 2024b; United Nations Development Programme and Open Government Products, 2024).

The AIBots platform complements Pair by allowing public officers to create custom retrieval-augmented chatbots in fewer than fifteen minutes, integrating internal agency knowledge bases up to the Restricted/Sensitive (Normal) classification level (Government Technology Agency of Singapore, 2025c). As of February 2025, AIBots had been used by approximately forty thousand officers across one hundred and fifteen agencies, who collectively created twelve thousand bots and sent over one million messages (Government Technology Agency of Singapore, 2025c). LaunchPad, the whole-of-government innovation platform launched in early 2023, hosts ideathons, prototypes, and central products and serves over three thousand monthly active users across all government agencies (Government Technology Agency of Singapore, 2025e; Microsoft Customer Stories, 2024; Civil Service College Singapore, 2024).

4.2 Citizen-Facing Services

The Singapore government has long deployed virtual assistants on agency websites. Ask Jamie, the cross-agency virtual assistant introduced in 2014, served more than ninety agencies at its peak and answered

over fifteen million queries ([Government Technology Agency of Singapore, 2020](#)). The Housing and Development Board (HDB) chatbot AskJudy provides instant replies to enquiries on HDB's main digital storefront, which itself receives more than two million views a month ([Housing Development Board, 2019](#)). Generative AI has also been used to develop new public-facing services, such as the National Library Board's ChatBook prototype that allows users to converse with books ([Ministry of Digital Development and Information, 2024](#)).

4.3 Sector-Specific Applications

Beyond cross-government platforms, agencies have deployed generative AI in sector-specific contexts spanning healthcare, the built environment, and clinical research, leveraging Pair Custom Assistants and AIBots to embed retrieval-augmented assistance into agency-specific workflows ([Government Technology Agency of Singapore, 2024b, 2025c](#); [United Nations Development Programme and Open Government Products, 2024](#)). The AI Trailblazers initiative, run jointly by the Ministry of Communications and Information, Digital Industry Singapore, the Smart Nation Group, and Google Cloud, addressed over one hundred generative AI use cases across eighty-four organisations in its first iteration ([Enterprise Singapore et al., 2024](#); [Computer Weekly, 2024](#)). Through the AI Government Cloud Cluster, public agencies have access to dedicated innovation sandboxes for prototyping generative AI solutions, and the more recent partnership with Google Cloud on Distributed Cloud Air-Gapped infrastructure enables agencies to deploy agentic AI while keeping sensitive data within on-premises data centres disconnected from the public internet ([GovInsider, 2025](#)).

4.4 Cross-Cutting Observations

These deployments span a wide range of risk profiles. Internal productivity tools such as Pair Chat operate primarily as drafting aids with strong human oversight and limited autonomy, while citizen-facing chatbots interact directly with members of the public on consequential domains including housing, taxation, and social support. A unified observability framework must therefore scale across at least an order of magnitude in risk and operational consequence. It must require enough telemetry from high-risk public-facing deployments to support audit and incident response, while not imposing the same overhead on low-stakes internal pilots where the benefits of formal monitoring are smaller and the ability to absorb the cost is lower.

5 Challenges and Considerations Unique to Public Sector Deployment

5.1 Accountability and Public Trust

Public sector services operate under an implicit accountability contract with citizens. When a government chatbot gives advice on housing eligibility, a citizen who acts on that advice has a legitimate expectation that the answer was correct, that the system was authorised to provide it, and that there is a record of what was said ([Leong, 2024](#)). Failures of public sector AI are not merely customer-experience events; they can affect rights, entitlements, access to services, and public trust in the institutions that deploy them. Empirical work on public sector AI adoption has consistently identified trust as the defining barrier, with research showing that nearly half of the public are unsure whether AI will be used safely in government services ([Luterek et al., 2025](#)). Observability is the substrate that makes this accountability concrete: without detailed logs of what the system did and why, agencies cannot demonstrate due diligence or explain decisions after the fact.

5.2 Regulatory and Policy Constraints

Public sector deployments operate within a denser web of legal and policy constraints than most commercial systems. In Singapore, the Personal Data Protection Act 2012 sets out obligations on the collection, use, and disclosure of personal data, including consent, purpose limitation, accuracy, retention limitation,

and accountability ([Government of Singapore, 2012](#)). Although public agencies are subject to a separate set of data-handling rules under the Public Sector (Governance) Act, the practical expectation is that government AI systems should match or exceed the protections offered to citizens by the PDPA. The PDPC's 2024 Advisory Guidelines on the Use of Personal Data in AI Recommendation and Decision Systems ([Personal Data Protection Commission, 2024](#)) extend this expectation to AI specifically, articulating principles around data minimisation, transparency, and accountability that apply throughout the AI lifecycle.

Information classification regimes add a further layer of constraint. Singapore's classification scheme (Restricted, Confidential, Secret) determines which infrastructure can host particular data, which models can process it, and which knowledge bases can be surfaced. AIBots, for example, is approved for use up to the Restricted/Sensitive (Normal) classification ([Government Technology Agency of Singapore, 2025c](#)), and higher-classification workloads require infrastructure such as the Government Distributed Cloud Air-Gapped environment ([GovInsider, 2025](#)). The presence of multiple classification tiers has direct consequences for observability: telemetry from a high-classification system cannot be exported to a low-classification observability backend, and any tool used for monitoring must conform to the security posture of the system it monitors.

5.3 Data Sensitivity and Sovereignty

Many agency use cases involve personally identifiable information or operationally sensitive data. The need to keep this data within sovereign or on-premises infrastructure rules out certain SaaS observability options and privileges open-source or self-hosted tools. Singapore's recent partnership with Google Cloud to deploy agentic AI on Distributed Cloud Air-Gapped infrastructure makes this explicit, with public officers able to develop and deploy AI agents while keeping sensitive data within on-premises data centres that are fully disconnected from the public internet ([GovInsider, 2025](#)). Observability telemetry inherits the same constraints as the data that produces it, which means that any whole-of-government observability standard must be implementable on infrastructure with strict data-residency rules.

5.4 Operational Heterogeneity

Agencies differ widely in technical maturity, in-house engineering capacity, and existing observability stacks. Some agencies have established Site Reliability Engineering practices, mature distributed-tracing backends, and dedicated AI operations teams; others are building their first generative AI deployment with a small team and limited tooling. Empirical research on public sector AI adoption has identified capability gaps and skills shortages as among the most consistent barriers to AI uptake ([Luterek et al., 2025](#); [Perdana and Mokhtar, 2025](#)). A viable framework must therefore be adoptable by agencies operating at very different levels of readiness, and must avoid prescribing a single technical stack. The Observability Requirements Framework introduced in this paper addresses this heterogeneity by tiering requirements by risk, so that mandatory items remain feasible for agencies with limited engineering capacity.

5.5 Agentic-AI-Specific Risks

Autonomous tool use introduces new risk surfaces that the public sector cannot afford to ignore. The ARC framework documents three categories of risk for agentic systems: component risks arising from the choice of LLM, tools, instructions, and memory; design risks arising from agentic architecture, access controls, and monitoring; and capability risks arising from the actions the agent can autonomously execute, such as running code, searching the internet, or modifying documents ([Khoo et al., 2025](#)). Several of these risks have been demonstrated empirically. Prompt injection attacks have been ranked the top threat to LLM applications by OWASP ([OWASP Foundation, 2025](#)), and recent benchmarks have shown that LLM agents are vulnerable to prompt injection attacks delivered through tool outputs, with attack success rates that remain non-trivial even against the strongest models ([Debenedetti et al., 2024](#)). Multi-agent systems introduce further risks: malicious instructions in one agent's context can propagate

silently through inter-agent communication and infect a multi-agent system as a whole (Lee and Tiwari, 2024). Recovery and rollback are harder when actions have been taken against external systems, which elevates the importance of end-to-end distributed traces and immutable audit logs.

5.6 Failure Modes Specific to Generative AI

Generative AI systems exhibit failure modes that pre-generative monitoring did not have to address. Hallucination, the production of plausible-sounding but factually incorrect or ungrounded statements, has been the subject of extensive recent research and remains a moving target even with retrieval augmentation and citation grounding (Huang et al., 2025). Silent regression after prompt or model updates is another characteristic failure mode, where a change that improves performance on one class of inputs degrades it on another without producing any error signal that traditional monitoring would detect. Bias and fairness gaps across demographic segments require dedicated telemetry that captures performance by sensitive attribute. Refusal events, where the system declines to answer, are a class of telemetry in their own right, because both excessive and insufficient refusal carry consequences for trust and utility. None of these failure modes is adequately covered by aggregate latency or error-rate metrics, and each requires explicit instrumentation.

6 The Importance of Observability in Public Sector LLMOps and AgentOps

6.1 Observability as the Governance Substrate

It is useful to distinguish monitoring from observability. Monitoring tracks known metrics against known thresholds and is appropriate when the failure modes of a system are well understood. Observability is the broader capability of being able to answer questions that were not anticipated at design time, by retaining sufficient telemetry to reconstruct system behaviour after the fact. Public sector generative AI requires the latter, because failure modes are still being discovered and each new deployment expands the surface that practitioners and auditors may need to reason about (Dong et al., 2024; OpenTelemetry Project, 2025a). Observability is also the artefact that connects high-level governance principles, such as accountability, transparency, fairness, and human oversight, to engineering practice. Each of these principles imposes a question that can only be answered if the underlying telemetry exists: an accountability audit needs prompts and responses with timestamps; a transparency report needs counts of refusals by category; a fairness assessment needs segment-wise performance metrics; a human-oversight review needs override logs.

6.2 Alignment with Existing Frameworks

The Singapore Model AI Governance Framework for Generative AI articulates accountability, transparency, explainability, and human oversight as core principles (Infocomm Media Development Authority and AI Verify Foundation, 2024). The Model AI Governance Framework for Agentic AI extends this orientation with four dimensions: assessing and bounding risks upfront, making humans meaningfully accountable, implementing technical controls and processes, and enabling end-user responsibility (Infocomm Media Development Authority, 2026). The ARC framework provides architectural guidance on how to implement these dimensions for agentic systems, including controls for monitoring and traceability, roles and access controls, and tamper-evident audit trails (Khoo et al., 2025; Government Technology Agency of Singapore, 2025d). Each of these frameworks calls for observability without specifying it in detail.

International frameworks converge on a similar direction. The NIST AI Risk Management Framework structures AI risk management around the four functions of Govern, Map, Measure, and Manage (National Institute of Standards and Technology, 2023), and its Generative AI Profile extends this structure with risks specific to generative AI such as confabulation, dangerous content, data privacy, and information integrity, paired with extensive lists of suggested mitigation actions across each risk category (National

Institute of Standards and Technology, 2024). The European Union AI Act imposes binding obligations on providers and deployers of high-risk AI systems, including automatic logging requirements under Article 12, post-market monitoring under Article 72, and human oversight requirements under Article 14 (European Parliament and Council of the European Union, 2024; Schuett, 2024). The OWASP Top 10 for LLM Applications enumerates the security risks specific to LLM deployments and ranks prompt injection as the foremost threat (OWASP Foundation, 2025). Together, these frameworks demand a similar set of observability capabilities, even as they differ in legal force and procedural detail.

6.3 The Current Gap

Despite this alignment, there is no shared specification for what telemetry agencies should collect, how to prioritise it under resource constraints, or how to define metrics consistently across deployments. The consequence is fragmented practice. One agency may log prompts and responses but not retrieval traces; another may capture tool calls but not subtask decompositions; a third may rely on aggregate user feedback without any structured incident logging. The results are difficult to compare across agencies and difficult to roll up to whole-of-government oversight. Open-source observability tooling has matured rapidly to meet the demand for LLM-specific telemetry, with projects such as Langfuse (Langfuse, 2024) and Arize Phoenix (Arize AI, 2024) now widely adopted and the OpenTelemetry community publishing semantic conventions for generative AI spans, events, and agent calls (OpenTelemetry Project, 2024, 2025a,b). These tools provide the substrate, but they do not by themselves answer the question of which telemetry an agency should be obliged to collect at which level of risk. The Observability Requirements Framework introduced in the next section is intended to fill this gap by providing a shared, risk-tiered specification that agencies can adopt and adapt to their local context.

7 The Observability Requirements Framework

7.1 Framework Goals and Non-Goals

The Observability Requirements Framework is designed with four explicit goals.

First, the framework provides a structured, risk-tiered specification of observability requirements for LLM-based and agentic systems in government, organised by operational concern, and intended as a portable artefact that can be adapted to jurisdictions other than Singapore. Second, it enables consistent implementation across agencies with differing technical maturity, by structuring requirements around a four-tier priority matrix that maps to system risk level and by pairing each requirement with a mathematical definition that anchors a comparable measurement across deployments. Third, it produces telemetry that is sufficient for the four functions that public sector deployments must support (e.g., debugging, incident investigation, safety auditing, and governance reporting) and the same telemetry creates the data substrate on which empirical research into deployed generative AI behaviour can be conducted across agencies and over time. Fourth, the framework aligns and operationalises the principles articulated in existing governance frameworks, in particular the Model AI Governance Framework for Generative AI (Infocomm Media Development Authority and AI Verify Foundation, 2024), the Model AI Governance Framework for Agentic AI (Infocomm Media Development Authority, 2026), and the ARC framework (Khoo et al., 2025; Government Technology Agency of Singapore, 2025d), and provides a worked example of the bridge between policy-level principles and engineering practice that comparable governance frameworks in other jurisdictions also need.

The framework is also explicit about what it does not do. The framework does not prescribe specific tools, vendors, or backend architectures, and it is intended to be implementable on a wide range of observability stacks including OpenTelemetry-based open-source pipelines (OpenTelemetry Project, 2025a), self-hosted tracing platforms such as Langfuse (Langfuse, 2024), and self-hosted evaluation platforms such as Arize Phoenix (Arize AI, 2024). The framework does not replace domain-specific evaluation, which remains the responsibility of each agency in light of its use case; rather, the framework specifies the substrate on which such evaluation can be built. Finally, the framework does not constitute

legal advice, and agencies should interpret its requirements alongside the binding obligations imposed by the Personal Data Protection Act ([Government of Singapore, 2012](#)) and any sector-specific regulations applicable to their deployment.

7.2 Adoption Flow

The framework defines a six-step flow that an agency team is intended to follow when applying it to a new or existing system. The flow is designed to fit naturally within the broader risk-and-control workflows that agencies already use under ARC ([Khoo et al., 2025](#)) and under whole-of-government AI policies.

1. **Evaluate key factors** before defining requirements. The team considers three groups of factors: the operational environment (domain criticality, user impact, integration with existing operations), the agent characteristics (autonomy level, task complexity, tool integration, learning capability), and the organisational constraints (regulatory requirements, agency requirements, existing infrastructure, technical expertise, risk tolerance).
2. **Determine the system risk level** as Low, Medium, or High, using the criteria set out in Section 7.3. Risk level, rather than data classification, is the primary determinant of which telemetry requirements are mandatory, although data classification interacts with risk in practice.
3. **Apply the Observability Requirements Framework** by reading the requirements table (Table 2) and adapting it to the system's risk level using the telemetry requirements matrix (Table 1).
4. **Phase implementation by priority**, starting with mandatory items and expanding as the system matures. Teams begin with offline batch evaluation before adding real-time online evaluation, and prioritise telemetry types that have direct downstream consumers (incident response, governance reporting).
5. **Validate regularly** by reviewing implementation against the requirements table and confirming that mandatory coverage is maintained as the system evolves. Validation is most effective when it is built into the agency's existing change-management process.
6. **Plan for evolution** by treating the framework as a living specification. As governance frameworks update and as the state of the art on LLM and agent observability advances, the metric definitions and requirement tiers will need to be revised.

7.3 Risk Tiering

The framework defines three risk tiers.

Low-Risk Operations include beta testing, internal pilots, and proofs of concept where the impact of failure is limited and human-in-the-loop oversight is strong. Internal experimentation by a single team to prototype a new use case, where outputs are not used to inform downstream decisions, falls into this tier.

Medium-Risk Operations include internal agency usage with minimal impact on operations, and low-impact public-facing use cases with strong human-in-the-loop oversight. An internal scoping assistant used by public officers to draft documents, with all outputs reviewed before use, is a representative example.

High-Risk Operations include systems with significant operational impact and high-impact public-facing deployments. Public-facing chatbots that answer questions on consequential domains such as housing, taxation, or social support, and agentic systems that take actions against external services on behalf of citizens, fall into this tier.

The framework treats risk tier as the primary determinant of mandatory telemetry, but it is important to recognise that data classification interacts with risk in practice. A low-risk system that processes high-classification data inherits stricter handling requirements for its telemetry, and a high-risk system that handles only public information may have stricter logging needs but laxer storage requirements. Agencies should therefore conduct a parallel data-handling assessment alongside the risk-tier determination.

7.4 Framework Components

The framework is constructed from two components: six operational categories that organise the telemetry requirements, and a four-tier priority structure expressed through the telemetry requirements matrix.

7.4.1 Operational Categories

The framework organises telemetry requirements into six categories that jointly cover the artefacts necessary to reconstruct, evaluate, and govern an agentic system.

1. **Agent-Level Observability.** Supports debugging of individual queries, provides visibility into agent decision pathways, surfaces anomalies in output, supports root cause analysis, and provides an audit trail for accountability.
2. **Agent Performance.** Tracks changes in input and output distributions, measures groundedness and factual accuracy, captures human oversight and intervention rates, and quantifies safety and fairness indicators.
3. **User Interaction and Behaviour.** Assesses user satisfaction and engagement, identifies common intents and usage patterns, surfaces UX pain points, detects abuse and misuse, and supports feedback-driven improvement.
4. **System Performance, Health, and Reliability.** Tracks availability, efficiency, regression, anomaly detection, SRE readiness, post-incident analysis, and SLA conformance.
5. **Safety and Security Compliance.** Monitors policy enforcement, boundary containment, responsible AI compliance, sensitive data handling, and threat detection.
6. **Product Lifecycle and Governance.** Captures the maturity and effectiveness of the agency's LLMOps processes, change and audit controls, and the cadence of structured review.

7.4.2 Telemetry Requirements Matrix

The matrix in Table 1 separates two ideas that are often conflated. The *tier* of a telemetry item is an intrinsic property of the item itself: it captures how foundational the item is to the framework, independent of any deployment. **Tier 1** items are the most foundational telemetry, anchoring routine production oversight, and include prompt and response logs, tool call logs, and basic error logs. **Tier 2** items extend the substrate to support incident investigation and accountability, including knowledge-base access logs, distributed traces, and override tracking. **Tier 3** items support deeper performance and safety analysis, such as drift detection, override-reason annotations, and segment-wise fairness measures. **Tier 4** items reflect research-grade or aspirational telemetry, such as hallucination rate on labelled samples and bias auditing on representative test sets. The *action* required of an agency in response to a tiered item then depends on the intersection of that tier with the system's risk level, as set out in the matrix.

The matrix uses three actions. *Mandatory* means the requirement must be implemented, with no waiver permitted. *Mandatory but Waivable* means the requirement must be implemented unless a waiver is obtained from the agency's responsible-AI oversight function with a documented risk acceptance. *Good-to-Have* means the requirement is strongly encouraged but not required, and agencies may implement it based on their specific needs and constraints. The full mapping is set out in Table 1: Tier 1 items are *Mandatory* for High- and Medium-Risk systems and *Good-to-Have* for Low-Risk systems. Tier 2 items are *Mandatory* for High-Risk systems, *Mandatory but Waivable* for Medium-Risk systems, and *Good-to-Have* for Low-Risk systems. Tier 3 items are *Mandatory but Waivable* for High-Risk systems and *Good-to-Have* for Medium- and Low-Risk systems. Tier 4 items are *Good-to-Have* for all systems. The matrix is deliberately proportional: as risk decreases, the action attaches one step lower on the same item, so that low-risk pilots are not held to the same evidentiary standard as high-impact citizen-facing deployments.

Table 1: Telemetry Requirements Matrix.

Tier	High-Risk System	Medium-Risk System	Low-Risk System
① Tier 1	Mandatory	Mandatory	Good-to-have
② Tier 2	Mandatory	Mandatory but Waivable	Good-to-have
③ Tier 3	Mandatory but Waivable	Good-to-have	Good-to-have
④ Tier 4	Good-to-have	Good-to-have	Good-to-have

7.4.3 Terminology: Session, Query, Task, and Request

Several telemetry items refer to a hierarchy of execution units that the framework defines explicitly. A **query** is a user-issued natural language prompt or input to the agentic system that initiates a session or a task. A **task** is a discrete goal that the agent attempts to fulfil in response to a query, which may require multiple reasoning steps, subtasks, or tool invocations. A **request** is a specific operation or API call performed by the agent during a task; multiple requests make up the actions of a single task. A **session** is a complete unit of interaction between the user and the system, encompassing one or more queries and all downstream agent tasks and requests. This hierarchy is consistent with the OpenTelemetry GenAI agent semantic conventions, which similarly distinguish tasks, actions, and agent invocations ([OpenTelemetry Project, 2025b,a](#)).

7.5 The Observability Requirements Table

Table 2 presents the full Observability Requirements Table, organised by operational category. Each row identifies a risk (what could go wrong), an impact statement (why it matters), and a set of controls combining mitigations with the metrics, logs, and traces that operationalise them. Each metric, log, or trace entry is annotated with a priority tier (①, ②, ③, or ④) that maps to the four-tier matrix in Table 1. The mapping between risk and control is intended to make explicit why each telemetry item exists, so that agencies can justify both implementation and waiver decisions in audit settings.

Table 2: Observability Requirements Table. Priority-tier markers ①–④ map to the four-tier matrix in Table 1.

Category	Risk (What Could Go Wrong)	Impact (Why It Matters)	Controls (Mitigations and Metrics / Logs / Traces)
Agent-Level Observability Goals: Support debugging of individual queries; provide visibility into agent decision pathways; surface anomalies in output; support root-cause analysis; provide an audit trail for accountability.	Agent actions and decision steps are not logged in sufficient detail to reconstruct specific sessions when necessary.	Cannot perform root-cause analysis or incident investigation. Fails the ARC framework’s baseline requirements for monitoring and traceability and for tamper-evident audit trails.	Mitigations: Centralised, access-controlled logging pipeline; audit logs for prompts, responses, and tool invocations; log retention and access policies aligned with organisational privacy and regulatory requirements; alerting when logging or tracing drops below expected coverage (e.g., missing trace IDs). Metrics / Logs / Traces: ① LLM prompt and response pairs with intermediate reasoning steps (where available), timestamps, and token usage. ① Tool / API call events with parameters, outcomes, and retries. ② Knowledge base / memory access logs (queries, retrieved documents, memory writes). ② Unique trace IDs per query–task–tool call chain (end-to-end distributed tracing). ① Error and exception logs. ② Subtask decomposition. ② Retrieval-to-response traces.

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Category	Risk	Impact	Controls
	Runaway or looping behaviour by the agent (e.g., infinite tool calls, repeated failures).	Resource exhaustion and potential denial of service. The agent may repeatedly hit external systems or tools with harmful or unintended actions.	<i>Mitigations:</i> Circuit breakers that stop tasks exceeding maximum steps, latency, or anomalous tool usage (aligned with ARC's architecture controls); guardrails in agent logic that cap loops and retries; automated alerts when loop patterns are detected, with auto-handover to a human if breached. <i>Metrics / Logs / Traces:</i> ① Tool usage count per task/session with thresholds for abnormal spikes. ① Task duration and step count per task/session. ② Error lineage traces linking repeated failures. ① Request retry counts per tool call.
	Silent regressions after model, tool, or prompt changes.	Behaviour changes are not detected until after harm or a major incident occurs. Violates governance expectations on controlled rollout and monitoring.	<i>Mitigations:</i> Change-management process with approvals for high-risk changes; canary or shadow deployments with explicit monitoring of safety metrics; automatic rollback triggers when key safety indicators degrade beyond thresholds. <i>Metrics / Logs / Traces:</i> ② Model deployment / change logs (who, what, when). ② Configuration / prompt / policy change logs with approver identity. ④ Side-by-side performance metrics pre- and post-change (hallucination rate, refusal rate, etc.).
Agent Performance Goals: Track changes (and drifts) in agent input/output distributions and performance over time; assess groundedness and factual accuracy; track level of human oversight and intervention; quantify safety and fairness indicators to identify harmful or biased outputs.	Undetected model performance drift on critical tasks or domains.	Gradual increase in harmful or incorrect outputs.	<i>Mitigations:</i> Periodic evaluation on safety-critical test sets, with results logged and compared over time; human-in-the-loop workflows for high-impact actions; investigations triggered when override rate crosses a threshold for specific flows. <i>Metrics / Logs / Traces:</i> ③ Performance / model-drift metrics on critical tasks (context- and agency-specific). ② Human override / edit logs and counts, especially for harmful or incorrect outputs. ④ Task completion rate per session, with explicit labelling of <i>safe failure</i> (the agent declines and the task is not completed) versus <i>unsafe success</i> (the task is completed but the agent emits unsafe or policy-violating content).
	Hallucinations or ungrounded responses leading to harmful advice (e.g., policy advice).	Legal liability and reputational damage.	<i>Mitigations:</i> Require citations or evidence for high-risk responses, with logging when absent; safety policy layer or secondary-checker model for high-risk content; tight domain scoping with refusal of out-of-domain tasks; periodic manual audits of random output samples with incident logging. <i>Metrics / Logs / Traces:</i> ④ Hallucination rate on sampled outputs (number of materially incorrect answers). ④ Number of ungrounded responses (no sources or weak citations) for high-risk domains. ③ Retrieval metrics (e.g., top-K recall, retrieval accuracy, embedding drift) where retrieval is supposed to ground responses.

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Category	Risk	Impact	Controls
	Biased or unfair outputs across demographic-related inputs.	Discrimination, regulatory compliance, ethics breaches.	<i>Mitigations:</i> Regular fairness audits with logged findings and remediation plans; updates to prompts, filters, or training data to reduce observed bias; human review for sensitive use cases or impacted groups. <i>Metrics / Logs / Traces:</i> ④ Bias and fairness indicators (e.g., parity metrics) on representative test sets. ④ Segment-wise hallucination or refusal rates by sensitive attribute.
User Interaction & Behaviour <i>Goals:</i> Assess user satisfaction and engagement; surface common intents, preferences, and usage patterns; identify UX pain points and unmet needs; detect abuse or misuse via abnormal usage patterns; identify prompts or topics that receive negative feedback; identify features most used versus those the service fails to handle.	Abusive or malicious user behaviour (e.g., prompt injection, scams, probing for sensitive data) goes undetected.	Security compromise, data leakage, and policy violations. The ARC “monitoring and traceability” risk is not properly mitigated.	<i>Mitigations:</i> Rate-limiting, IP/user throttling, and blocking for abusive activity; input validation and sanitisation; strong filters against prompt injection; rules or prompts that constrain model behaviour in the presence of adversarial input. <i>Metrics / Logs / Traces:</i> ② Logs of detected unexpected or out-of-scope query types and topics. ① Content policy violation attempts and model refusals per user/session. ① Session traces for flagged abusive sessions.
	High rate of human handover or override but no structured tracking.	Hidden failure modes and an inability to demonstrate to regulators that safety controls actually work.	<i>Mitigations:</i> Explicit handover protocol with guidelines for operators; periodic review of high-risk handover cases for systemic fixes; training for operators to consistently label safety versus non-safety escalations. <i>Metrics / Logs / Traces:</i> ② Number of sessions requiring human handover or override. ③ Human annotations on the reason for handover (safety, quality, capability gap, policy issue). ② User feedback (thumbs up/down) tied to specific responses. ③ Repeat queries / user retry rates (number of repeated queries per session). ③ Number of users abandoning their sessions.

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Category	Risk	Impact	Controls
System Performance, Health & Reliability Goals: Track reliability and availability over time; measure operational efficiency and cost-effectiveness; detect regressions early; enable anomaly detection on system-level metrics; support SRE practice with alerts and dashboards and human intervention as needed; facilitate post-incident root-cause analysis; ensure conformance with SLAs and budget constraints.	System outages, time-outs, or partial failures that produce unsafe states (e.g., an external tool action completed but its response is lost).	Incomplete operations and inconsistent state. Loss of traceability for some actions.	Mitigations: SRE-style monitoring with on-call rotation for safety-critical systems; graceful degradation and safe-fail behaviours (e.g., retry rather than incomplete execution); transactional semantics or idempotency for critical tools; failover and redundancy for critical dependencies. Metrics / Logs / Traces: ② System uptime / availability. ② Error rates (e.g., HTTP 5xx, tool invocation failures). ② Timeout / abort counts per request and per tool. ① System request logs. ① System timeout / abort logs. ① System error logs and crash reports. ② System throughput and queue length. ② GPU / CPU / memory utilisation (for capacity risk). ② Distributed request traces across services (where multi-service flows exist). ③ Redundancy and failover success rates.
	Unexpected traffic spikes or abuse.	Resource exhaustion and degraded safety controls. Monitoring, filtering, and policy enforcement components may fail under load.	Mitigations: Auto-scaling with limits specifically for safety-critical components (e.g., filters, policy engines, logging pipeline); rate-limiting and quota enforcement at API gateways; load testing under adversarial traffic patterns. Metrics / Logs / Traces: ② Traffic volume per user. ② Error and latency distribution during spikes. ③ Cost per query, task, or session.
Safety & Security Compliance Goals: Monitor and enforce ethical guidelines, laws, and government policies; ensure agent behaviour stays within defined use boundaries; build trust and satisfy Responsible AI requirements and reporting; ensure sensitive or protected data is handled properly and not leaked; surface potential security threats (e.g., prompt injection attempts, malicious inputs) by tracking anomalous events.	Harmful, toxic, or policy-violating outputs that are not detected or not logged.	Direct policy violations and potential legal breaches. Cannot demonstrate due diligence in post-incident review.	Mitigations: Strong content policy and safety filters on the user-facing interface; safety tuning and system prompts aligned to policy; clear escalation workflow for flagged incidents with human review and logging. Metrics / Logs / Traces: ③ Toxic / harmful output rate (flagged by filters or human review). ② Number of policy violations detected versus refused. ② Model refusal events and categories (e.g., safety, privacy). ③ Distribution of content categories requested (especially sensitive topics).

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Category	Risk	Impact	Controls
	Data privacy or confidentiality breaches (e.g., PII or sensitive organisational data exposed in outputs or logs).	Regulatory fines, contractual breaches, and reputational harm.	<i>Mitigations:</i> Data classification and tagging, with controls on what can be retrieved and surfaced; pseudonymisation or anonymisation of logs where feasible; encryption in transit and at rest; strict role-based access control on log access; periodic privacy audits of logs and training data. <i>Metrics / Logs / Traces:</i> ② Privacy incident count (PII exposure events). ② Privacy breach incident logs (records of detected personal or sensitive data leaks). ② Classified information leakage incidents. ② Memory / KB access logs with classification of data sensitivity.
	Unauthorised tool usage or privilege escalation by agents or users.	Fraud, data exfiltration, and abuse of external systems.	<i>Mitigations:</i> Strong roles and access controls for agents and tools (aligned with ARC's Roles & Access Controls); short-lived, scoped credentials; hardware or high-assurance authorisation for critical tools; automated anomaly detection on privileged tool usage patterns. <i>Metrics / Logs / Traces:</i> ② Unauthorised or failed privileged tool usage attempts. ② Audit logs of role / permission changes and token issuance. ② Agent identity / credential usage logs.
	Incomplete or mutable audit trails for safety incidents.	Cannot prove accountability or reconstruct events for regulators.	<i>Mitigations:</i> Append-only log stores or log encryption (per ARC guidance); retention policies that cover regulatory requirements; separation of duties so that operators cannot modify audit logs. <i>Metrics / Logs / Traces:</i> ② Full conversation traces retained for flagged safety and security incidents. ② Audit logs for prompts, responses, tool calls, and memory operations.
Product Lifecycle & Governance <i>Goals:</i> Provide insight into the maturity and effectiveness of the agency's LLMOps and AgentOps processes.	Model, prompt, or policy changes are made without audit, safety assessment, or rollback path.	Safety regressions slip into production. Weak governance and non-compliance with internal AI policies.	<i>Mitigations:</i> Governance process requiring risk and impact assessment before high-risk changes; structured safety review and sign-off for new capabilities; experiment tracking with safety metrics (hallucination rate, harmful output rate, bias indicators) as first-class KPIs. <i>Metrics / Logs / Traces:</i> ③ Deployment metrics: rollout frequency and percentage of changes with safety evaluation. ③ Audit metrics: percentage of audited sessions and number of audit findings per period. ② Configuration change logs with owners and approvers.
	Limited or ad hoc safety audits and incident post-mortems.	Repeated incidents and a perception by regulators that controls are ineffective.	<i>Mitigations:</i> Regular, documented safety reviews (e.g., quarterly) with tracked actions; post-incident review templates emphasising root cause, control gaps, and design changes; governance dashboards highlighting safety KPIs to leadership. <i>Metrics / Logs / Traces:</i> ③ Audit logs for reviewed sessions and findings. ③ Incident count and severity over time, with recurrence indicators.

7.6 Metric Definitions

The framework pairs Table 2 with mathematical definitions for each metric, log, and trace, so that any two agencies following the specification compute numerically comparable values. Table 3 provides definitions for every item in the requirements table, organised by operational category. The markers preceding each entry mirror those used in Table 2, indicating the item’s priority tier (①–④).

Notation

The definitions in Table 3 use a consistent set of mathematical conventions, summarised here. We write $\#X$ for the count of events or items of type X over the relevant collection scope; for example, $\#\text{refusals}$ denotes the count of refusal events, and $\#\{\text{outcome} = \ell\}$ denotes the count of sessions whose outcome equals label ℓ . Counts are always taken over an implicit scope that is named by the subscript: s for a session, T for a task, i for a turn within a session, g for a user segment or group, k for a retrieval event, and j for a tool-call event. A ratio of two counts is written $\frac{a}{b}$ and is defined when $b > 0$. A log is written as a set of tuples whose fields are named explicitly: $\text{log_name} = \{(\text{field}_1, \text{field}_2, \dots, \text{field}_n)\}$, with $|\text{log_name}|$ denoting its cardinality at the time of measurement. Time-averaged quantities use the bar notation $\bar{x} = \frac{1}{\Delta t} \int x(t) dt$, where Δt is the observation window. Probabilities $P(\cdot)$ are interpreted as empirical relative frequencies over the indicated sample. Drift between two distributions or two performance states is written $\text{diff}(\cdot, \cdot)$; the framework leaves the choice of divergence measure (Kullback–Leibler, Wasserstein, Population Stability Index, or a domain-specific equivalent) to the implementing agency, on the understanding that the choice is documented and held constant across measurements of the same metric. Trace and graph structures use the standard convention $G = (V, E)$ for a directed graph with vertex and edge sets; named subgraphs (retry-lineage graphs, retrieval-to-response graphs, task decomposition trees) carry the additional semantics stated at their definition.

Table 3: Mathematical definitions for the metrics, logs, and traces of the Observability Requirements Framework.

Metric / Log / Trace	Mathematical definition
<i>Category: Agent-Level Observability</i>	
① LLM prompt and response pairs with intermediate reasoning steps, timestamps, and token usage.	For each turn i , $\text{log } r_i = (\text{timestamp}_i, \text{prompt}_i, \text{response}_i, \text{tokens}_{\text{in},i}, \text{tokens}_{\text{out},i}, \text{intermediate_steps}_i)$.
① Tool / API call events with parameters, outcomes, and retries.	For each call event e_j , $\text{log } e_j = (\text{timestamp}_j, \text{tool}_j, \text{parameter}_j, \text{outcome}_j, \text{retry_attempt}_j)$.
② Knowledge base / memory access logs (queries, retrieved documents, memory writes).	For each access event e_k , $\text{log } e_k = (\text{timestamp}_k, \text{query}_k, \text{retrieved_docs}_k, \text{memory_write}_k)$.
② Unique trace IDs per query–task–tool call chain (end-to-end distributed tracing).	Assign trace IDs for all events e in a query→task→tool chain: $\text{trace_id}_e = \text{constant}$. Coverage: $\text{trace_coverage} = \frac{ \{e : \text{trace_id}_e \text{ present}\} }{ \{e\} }$.
① Error and exception logs.	$\text{error_log} = \{(\text{timestamp}, \text{trace_id}, \text{error_type}, \text{error_message}, \text{stack_hash})\}$; $\text{error_count} = \text{error_log} $ over the observation window.
② Subtask decomposition.	Represent a task as a tree D whose nodes are subtasks and whose edges capture parent–child relations. $\text{decomposition_size} = \text{nodes}(D) $.
② Retrieval-to-response traces.	Trace graph $G = (V, E)$ linking retrieved items d to response spans y ; edge $(d \rightarrow y)$ exists if d was used to generate y .
① Tool usage count per session with thresholds for abnormal spikes.	$\text{tool_count}_s = \#\text{tool calls in } s$. Alert if $\text{tool_count}_s > \text{threshold}$.

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Table 3 continued from previous page.

Metric / Log / Trace	Mathematical definition
① Task duration and step count per task.	$\text{task_duration}_T = \text{task_end}_T - \text{task_start}_T$; $\text{step_count}_T = \text{\#actions in } T$.
② Error lineage traces linking repeated failures.	Build a retry-lineage graph whose nodes are attempts and whose edges link attempt $k \rightarrow k+1$ when a failure triggers a retry.
① Request retry counts per tool call.	$\text{\#retries} = \text{\#attempts} - 1$; $\text{retry_rate} = \frac{\text{\#retries}}{\text{\#calls}}$.
② Model deployment / change logs (who, what, when).	$\text{deployment_log} = \{(\text{timestamp}, \text{actor}, \text{model_id}, \text{model_version}, \text{environment})\}$.
② Configuration / prompt / policy change logs with approver identity.	$\text{change_log} = \{(\text{timestamp}, \text{change_type}, \text{artifact_id}, \text{description}, \text{owner}, \text{approver})\}$.
④ Side-by-side performance metrics pre- and post-change.	For any metric m : $\Delta m = m_{\text{after}} - m_{\text{before}}$; $\text{relative change} = \frac{m_{\text{after}} - m_{\text{before}}}{m_{\text{before}}}$.
<i>Category: Agent Performance</i>	
③ Performance / model-drift metrics on critical tasks.	$\text{drift} = \text{diff}(\text{performance}_{\text{current}}, \text{performance}_{\text{baseline}})$.
② Human override / edit logs and counts.	$\text{edit_count}_s = \text{\#edited outputs in } s$; $\text{edit_rate}_s = \frac{\text{edit_count}_s}{\text{\#outputs in } s}$.
④ Task completion rate per session, with safe-failure vs unsafe-success labelling.	For each session, label its outcome $\text{outcome} \in \{\text{safe_failure}, \text{unsafe_failure}, \text{unsafe_success}, \text{safe_success}\}$. Report, for each label ℓ , $P(\text{outcome} = \ell) = \frac{\text{\#\{outcome} = \ell\}}{\text{\#outcomes}}$.
④ Hallucination rate on sampled outputs.	$\text{hallucination_rate} = \frac{\text{\#materially_incorrect_responses}}{\text{\#sampled_responses}}$.
④ Number of ungrounded responses for high-risk domains.	$\text{ungrounded_count} = \text{\#high-risk outputs with missing or weak citations}$; $\text{ungrounded_rate} = \frac{\text{ungrounded_count}}{\text{\#high-risk outputs}}$.
③ Retrieval metrics (e.g., top- K recall, retrieval accuracy, embedding drift).	$\text{drift} = \text{diff}(\text{performance}_{\text{current}}, \text{performance}_{\text{baseline}})$, where $\text{performance} \in \{\text{top-}K \text{ recall}, \text{retrieval accuracy}, \text{embedding drift}\}$.
④ Bias and fairness indicators (parity metrics) on representative test sets.	$\text{parity_gap} = \max_g (\text{performance}_g - \text{performance}_{\text{ref}})$, where $\text{performance} \in \{\text{approval rate}, \text{refusal rate}, \text{hallucination rate}\}$.
④ Segment-wise hallucination or refusal rates by sensitive attribute.	For group g : $\text{rate}_g = \frac{\text{\#hallucination/refusal events for } g}{\text{\#outputs for } g}$. Report $\max_g \text{rate}_g$.
<i>Category: User Interaction & Behaviour</i>	
② Logs of detected unexpected / out-of-scope query types and topics.	$\text{OOS_log} = \{(\text{timestamp}, \text{session_id}, \text{oos_label})\}$; $\text{oos_rate} = \frac{\text{\#out-of-scope queries}}{\text{\#queries}}$.
① Content policy violation attempts and model refusals per user/session.	Per user u or session s : $\text{violation_attempt_rate} = \frac{\text{\#attempts}}{\text{\#queries}}$; $\text{refusal_rate} = \frac{\text{\#refusals}}{\text{\#queries}}$.
① Session traces for flagged abusive sessions.	For flagged session s , store the ordered sequence of events $\{(\text{timestamp}_e, \text{prompt}_e, \text{tool_call}_e, \text{output}_e)\}$ for reconstruction.
② Number of sessions requiring human handover and override.	$\text{override_count} = \text{\#sessions requiring handover/override}$; $\text{override_rate} = \frac{\text{override_count}}{\text{\#sessions}}$.
③ Human annotations on the reason for handover.	$\text{handover_annotation} = \{(\text{timestamp}, \text{session_id}, \text{handover_reason}, \text{description})\}$, where $\text{handover_reason} \in \{\text{safety}, \text{quality}, \text{capability gap}, \text{policy}\}$.
② User feedback (thumbs up/down) tied to specific responses.	$\text{satisfaction_rate} = \frac{\text{\#thumbs up}}{\text{\#thumbs up} + \text{\#thumbs down}}$.
③ Repeat queries / user retry rates per session.	$\text{repeat_rate} = \frac{\text{\#repeated queries in session}}{\text{\#queries in session}}$.

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Metric / Log / Trace	Mathematical definition
③ Number of users abandoning their sessions.	$\text{abandon_rate} = \frac{\text{\#inactive sessions that timed out}}{\text{\#sessions started}}.$
<i>Category: System Performance, Health & Reliability</i>	
② System uptime / availability.	$\text{availability} = \frac{\text{total runtime} - \text{downtime}}{\text{total runtime}}.$
② Error rates (e.g., HTTP 5xx, tool invocation failures).	$\text{error_rate} = \frac{\text{\#errors}}{\text{\#requests}}.$
② Timeout / abort counts per request and tool.	Per request or tool: $\text{timeout_count} = \text{\#timeouts and aborts in the period}.$
① System request logs.	$\text{request_log} = \{(\text{timestamp}, \text{request_id}, \text{service_id}, \text{endpoint_id}, \text{status}, \text{latency_ms}, \text{bytes})\}.$
① System timeout / abort logs.	$\text{timeout_log} = \{(\text{timestamp}, \text{request_id}, \text{tool_id}, \text{timeout_reason})\}.$
① System error logs and crash reports.	$\text{error_log} = \{(\text{timestamp}, \text{service_id}, \text{error_code}, \text{stack_hash}, \text{crash_id})\};$ $\text{crash_count} = \text{\#system crashes in the period}.$
② System throughput and queue length.	$\text{throughput} = \frac{\text{\#requests}}{\Delta t}; \bar{q} = \frac{1}{\Delta t} \int q(t) dt,$ where Δt is the observation window and $q(t)$ is the number of requests waiting in the queue at time t .
② GPU / CPU / memory utilisation.	$\bar{u} = \frac{1}{\Delta t} \int \text{util}(t) dt,$ where Δt is the observation window and $\text{util}(t)$ is the utilisation of GPU / CPU / memory at time t .
② Distributed request traces across services.	trace is a set of spans across services; each span = $(\text{trace_id}, \text{span_id}, \text{parent_id}, \text{service_id}, \text{timestamp}_{\text{start}}, \text{timestamp}_{\text{end}}).$
③ Redundancy and failover success rates.	$\text{failover_success_rate} = \frac{\text{\#successful failovers}}{\text{\#failover attempts}}.$
② Traffic volume per user.	For user u : traffic_volume_u is the number of requests, or bytes transferred, per period.
② Error and latency distribution during spikes.	$\text{latency_quantiles} = \{q_p : P(\text{latency} \leq q_p) = p\} \text{ for } p \in \{0.5, 0.95, 0.99\}.$
③ Cost per query, task, or session.	$\overline{\text{cost}} = \frac{\text{total cost}}{\text{\#queries / tasks / sessions}}.$
<i>Category: Safety & Security Compliance</i>	
③ Toxic / harmful output rate.	$\text{toxic_rate} = \frac{\text{\#outputs flagged by guardrails / human review}}{\text{\#outputs}}.$
② Number of policy violations detected versus refused.	$\text{refused_share} = \frac{\text{\#refusals due to policy violations}}{\text{\#policy violations}} \text{ per period}.$
② Model refusal events and categories.	For category c : $\text{refusal_count}_c = \text{\#refusals attributable to } c$. Report the distribution $\frac{\text{refusal_count}_c}{\sum_c \text{refusal_count}_c}$ across all categories.
③ Distribution of content categories requested.	$\text{distribution}(c) = \frac{\text{\#requests of category } c}{\text{\#requests}}.$
② Privacy incident count (PII exposure events).	$\text{privacy_incident_count} = \text{\#PII exposure events detected in the period}.$
② Privacy breach incident logs.	$\text{privacy_incident_log} = \{(\text{timestamp}, \text{incident_id}, \text{exposure_type}, \text{severity_level}, \text{trace_id})\}.$
② Classified information leakage incidents.	$\text{classified_leak_count} = \text{\#classified-data exposure events detected in the period}.$
② Memory / KB access logs with sensitivity classification.	For KB / memory accesses: $\text{access_log} = \{(\text{timestamp}, \text{agent_id}, \text{resource_id}, \text{sensitivity_level})\}.$
② Unauthorised or failed privileged tool-usage attempts.	$\text{unauthorised_rate} = \frac{\text{\#unauthorised privileged attempts}}{\text{\#privileged attempts}}.$

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Table 3 continued from previous page.

Metric / Log / Trace	Mathematical definition
② Audit logs of role / permission changes and token issuance.	$\text{authorised_change_log} = \{(\text{timestamp}, \text{actor}, \text{role/permission change}, \text{target principal})\}.$
② Agent identity / credential usage logs.	$\text{credential_use_log} = \{(\text{timestamp}, \text{agent_id}, \text{credential_id}, \text{scope}, \text{tool_id})\}.$
② Full conversation traces for flagged safety / security incidents.	For each flagged incident i , retain the full transcript $c_i = (\text{timestamp}, \text{messages}, \text{tool_id}, \text{agent_id})$ as an immutable record.
② Audit logs for prompts, responses, tool calls, memory operations.	With append-only integrity controls: $\text{audit_log} = \{(\text{timestamp}, \text{session_id}, \text{prompt}, \text{response}, \text{tool_id}, \text{memory_ops})\}.$
<i>Category: Product Lifecycle & Governance</i>	
③ Deployment metrics: rollout frequency, percentage of changes with safety evaluation.	$\text{rollout_frequency} = \frac{\text{\#deployments}}{\Delta t}; \text{ safety_evaluation_percent} = \frac{\Delta t}{\text{\#changes with safety evaluation}}.$
③ Audit metrics: percentage of audited sessions, audit findings per period.	$\text{audited_session_percent} = \frac{\text{\#audited sessions}}{\text{\#sessions}}; \text{ findings_rate} = \frac{\text{\#audit findings}}{\Delta t}.$
② Configuration change logs with owners and approvers.	$\text{change_log} = \{(\text{timestamp}, \text{change_type}, \text{artifact_id}, \text{description}, \text{owner}, \text{approver})\}.$
③ Audit logs for reviewed sessions and findings.	$\text{review_log} = \{(\text{timestamp}, \text{session_id}, \text{reviewer}, \text{finding_label}, \text{description}, \text{severity})\}; \text{ findings_count} = \text{\#findings in the period}.$
③ Incident count and severity over time, with recurrence indicators.	$\text{incident_count} = \text{\#incidents in the period}; \text{ recurrence_rate} = \frac{\text{\#recurrences}}{\text{\#incidents}}.$

7.7 Implementation Strategy

We recommend that agencies running Medium- or High-Risk systems adopt the framework through three implementation phases; agencies running Low-Risk pilots are not expected to follow the full phasing, but are encouraged to adopt Tier 1 items as good practice and to revisit the matrix when the system progresses toward production. **Phase 1** concentrates on Tier 1 items and stands up offline batch evaluation. This phase establishes the trace and log substrate (prompts, responses, tool calls, error logs, refusals, system telemetry) and ensures that data flows are captured before any analysis is layered on top. **Phase 2** adds Tier 2 items and incident-investigation workflows, including knowledge-base access logs, distributed request traces, override and handover tracking, and privacy-incident logging. **Phase 3** introduces the Tier 3 and Tier 4 items that require more sophisticated infrastructure, including drift metrics, hallucination evaluation, fairness indicators, and bias auditing on representative test sets. Within this phasing, agencies should lean on existing instrumentation where it exists, including OpenTelemetry-based distributed tracing for system telemetry ([OpenTelemetry Project, 2024, 2025a](#)), structured logging pipelines for security telemetry, and LLM-specific observability platforms such as Langfuse ([Langfuse, 2024](#)) or Arize Phoenix ([Arize AI, 2024](#)) for prompt-response and trace logging. At each phase, the action attached to each item (Mandatory, Mandatory but Waivable, or Good-to-Have) is read from Table 1 for the agency's risk level; the phasing recommends a sequence in which to address the items, while the matrix determines what is required of the agency at each step.

8 Concluding Remarks and Outlook

This manuscript has set out the rationale for, and the structure of, an Observability Requirements Framework for generative AI systems in the Singapore public sector. We argued that public sector

deployments of generative AI face an observability gap that cannot be closed by retrofitting earlier MLOps practice, and that this gap matters because observability is the substrate that connects governance principles to engineering practice. We situated the framework within Singapore's National AI Strategy 2.0, the Smart Nation 2.0 vision, and the trio of governance frameworks now in place at the national level: the Model AI Governance Framework for Generative AI, the Model AI Governance Framework for Agentic AI, and the Agentic Risk and Capability framework. We surveyed publicly disclosed Singapore government use cases, examined the considerations that distinguish public sector deployment from commercial practice, and presented the framework in detail with its goals, adoption flow, components, and supporting tables.

While the framework presented is necessary, it is not sufficient in isolation. Two further bodies of work are required to make the framework operational at whole-of-government scale, and these will be the subject of a companion manuscript currently in preparation.

The companion manuscript will, first, demonstrate the framework on two worked examples drawn from Singapore public sector deployments. The first example will apply the framework to a public-facing HDB resale transaction chatbot, taking a high-risk system through the six-step adoption flow and showing how each operational category maps onto concrete telemetry in a citizen-facing context. The second example will apply the framework to SailorMate, a multi-agent scoping assistant developed within SkillsFuture Singapore (SSG), illustrating how the framework scales to medium-risk internal deployments where the concerns shift from public-facing safety to inter-agent coordination, memory access, and subtask traceability. Together, the two examples will show that the framework accommodates both ends of the public sector deployment spectrum without requiring substantial customisation.

The companion manuscript will also examine the institutional moves required to make the framework durable. The first such move is integration of the framework into central platform products, including Pair, AIBots, and PlatformAI. Building Tier 1 and Tier 2 telemetry into these platforms by construction would allow agencies that build on them to inherit observability rather than implement it from scratch. The second move is encoding the framework as part of the policy requirements for future government generative AI products, either as part of existing AI governance guidance issued by IMDA, PDPC, and GovTech, or as a new technical circular issued through the Smart Nation and Digital Government Group. Policy encoding raises questions about scope, waiver processes, audit rights, and the cadence of revision that the framework itself does not address, and the companion manuscript will examine these questions in detail.

The trajectory we are describing is one in which observability becomes a default property of government generative AI, rather than a discretionary feature whose presence depends on the technical sophistication of the agency that deployed the system. The framework introduced in this paper is a step on that trajectory. The work that remains, demonstrated in the companion manuscript and refined through implementation across agencies, is to make the framework so well-integrated into platforms and policies that an agency cannot deploy a generative AI system without inheriting it.

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